

UCT41042 – Practical for Introduction to Smart System
Department of Information & Communication Technology
Faculty of Technology
South Eastern University of Sri Lanka
Lab Sheet No: 07

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Title: Temperature-Based Servo Motor Control with Manual Override using ESP32 & Blynk.

Aims:

To design and implement a temperature-based servo motor control system using the ESP32 microcontroller, where the servo motor operates automatically according to temperature variations, while also providing a manual override feature through the Blynk mobile application for real-time user control and monitoring.

Blynk IoT Platform

Blynk is a cloud-based IoT platform that enables real-time bi-directional communication between microcontrollers and smartphone apps. It uses virtual pins (V0–V255) as data channels. The Blynk library abstracts WebSocket/HTTP calls, allowing the developer to read/write virtual pins with simple function calls.

<https://blynk.io>

Task 01:

1. Project Setup in Wokwi / VS Code
2. Circuit Design
 - a. Connect the DHT sensor to an ESP32 GPIO pin
 - b. Connect the servo motor to a PWM-capable GPIO pin
 - c. Connect the LCD (I2C) to the ESP32.
3. Library Integration
 - i. Blynk
 - ii. DHT sensor library
 - iii. Servo
 - iv. LiquidCrystal_I2C

4. Blynk Configuration

- a. Create a New Template in Blynk (IoT platform)
- b. Assign virtual pins
 - i. Temperature: Units: Degrees, °
 - ii. Humidity: Units: Percentage, % | Max: 100
 - iii. Servo: Units: Degrees, ° | Max: 180
- c. Create a New Device
- d. Add widgets:
 - i. Temperature display (Gauge)
 - ii. Humidity display (Gauge)
 - iii. Servo (Switch)
- e. Obtain Template ID, Template Name and Auth Token

5. Programming Logic Development in Wokwi / VS Code

- a. Initialize Wi-Fi and Blynk connection
- b. Read temperature and humidity from DHT sensor
- c. Display the values in LCD

6. Simulation & Testing in Wokwi / VS Code

- a. Run simulation
- b. Adjust temperature values in Wokwi / VS Code
- c. Observe servo movement
- d. Verify Blynk interaction

Task 02: Implement Auto and Manual Mode for Servo Motor Control

Modify the existing temperature-based servo motor control project to support two operating modes:

- Auto Mode – Servo motor operates automatically based on temperature conditions.
- Manual Mode – Servo motor is controlled manually by the user using the Blynk.

Task Description

Students are required to enhance the existing system by introducing a mode selection mechanism using the Blynk IoT platform.

Requirements

1. Add an Auto/Manual Mode Switch in Blynk
 - a. Create a new Button (Switch widget) in Blynk.
 - b. Assign it to a new virtual pin (example: V3).
 - c. This switch will be used to select:
 - i. ON → Auto Mode
 - ii. OFF → Manual Mode
2. Auto Mode Functionality
 - a. When Auto Mode is enabled:
 - i. The servo motor should rotate automatically based on the temperature value read from the DHT sensor.
 - ii. Students must define a temperature condition, for example:
 1. If temperature ≥ 30 °C → rotate servo to 90°
 2. If temperature < 30 °C → rotate servo back to 0°
 - iii. Manual control from the servo switch should be disabled in this mode.
3. Manual Mode Functionality
 - a. When Manual Mode is enabled:
 - i. The servo motor should be controlled only using the existing Blynk servo control switch (Virtual Pin V2).
 - ii. Temperature-based automatic control must be ignored.
4. Programming Logic
 - a. Use a variable (for example, bool autoMode) to store the current mode.
 - b. Use BLYNK_WRITE() functions to:
 - i. Detect mode selection (Auto/Manual).
 - ii. Control servo movement based on the selected mode.
 - c. Ensure that:
 - i. Auto mode logic runs only when auto mode is active.
 - ii. Manual mode logic runs only when auto mode is inactive.
5. LCD Display
 - a. Display the current mode on the LCD:

Example:

 - Mode: AUTO
 - Mode: MANUAL